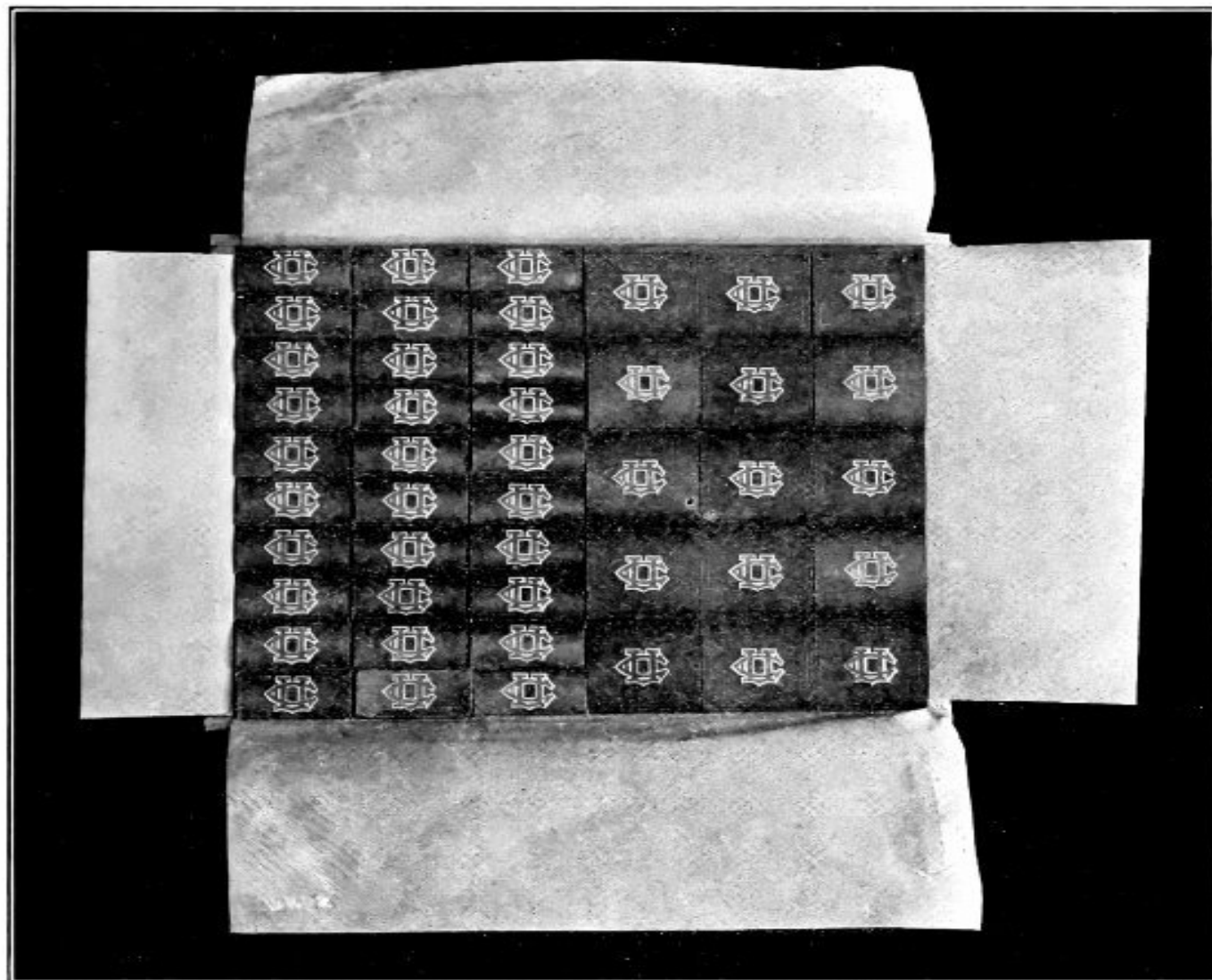


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COMPARISON OF WOODS FOR BUTTER BOXES

By G. D. TURNBOW



Proper method of packing cartoned butter in 60-pound boxes

BULLETIN No. 369

AUGUST, 1923

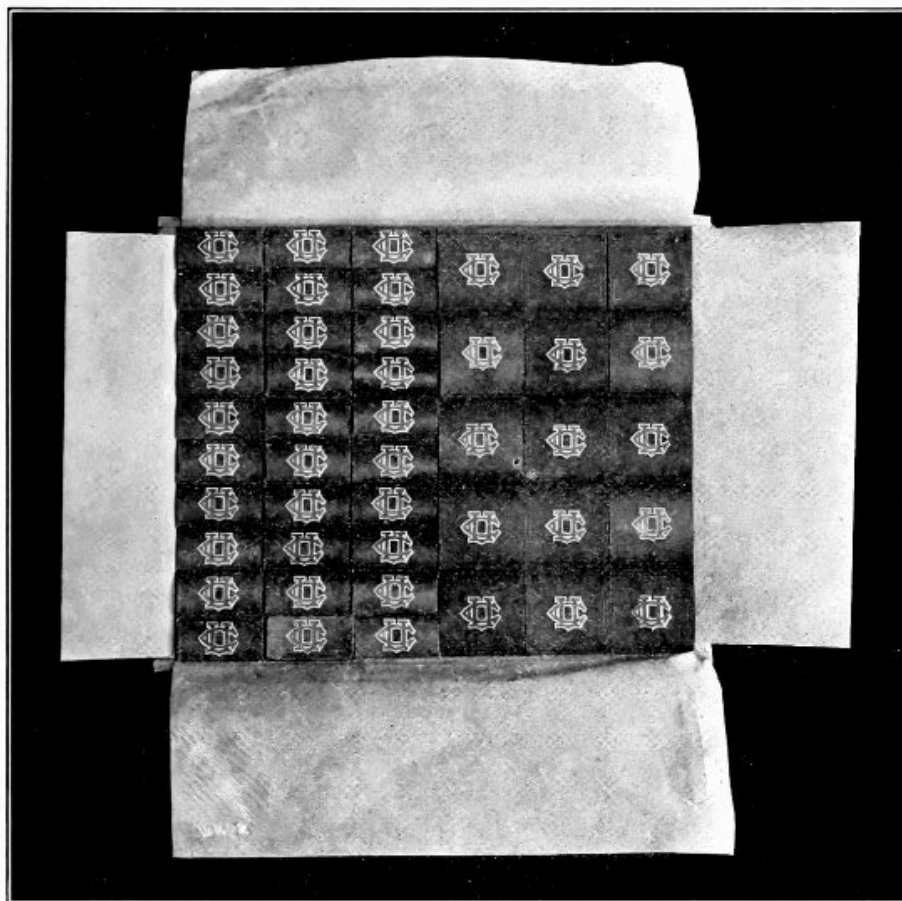
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Title: Comparison of Woods for Butter Boxes

Author: Grover Dean Turnbow

Release date: February 8, 2022 [eBook #67359]

Most recently updated: October 18, 2024

Language: English

Original publication: United States: University of California Press, 1923

Other information and formats: www.gutenberg.org/ebooks/67359

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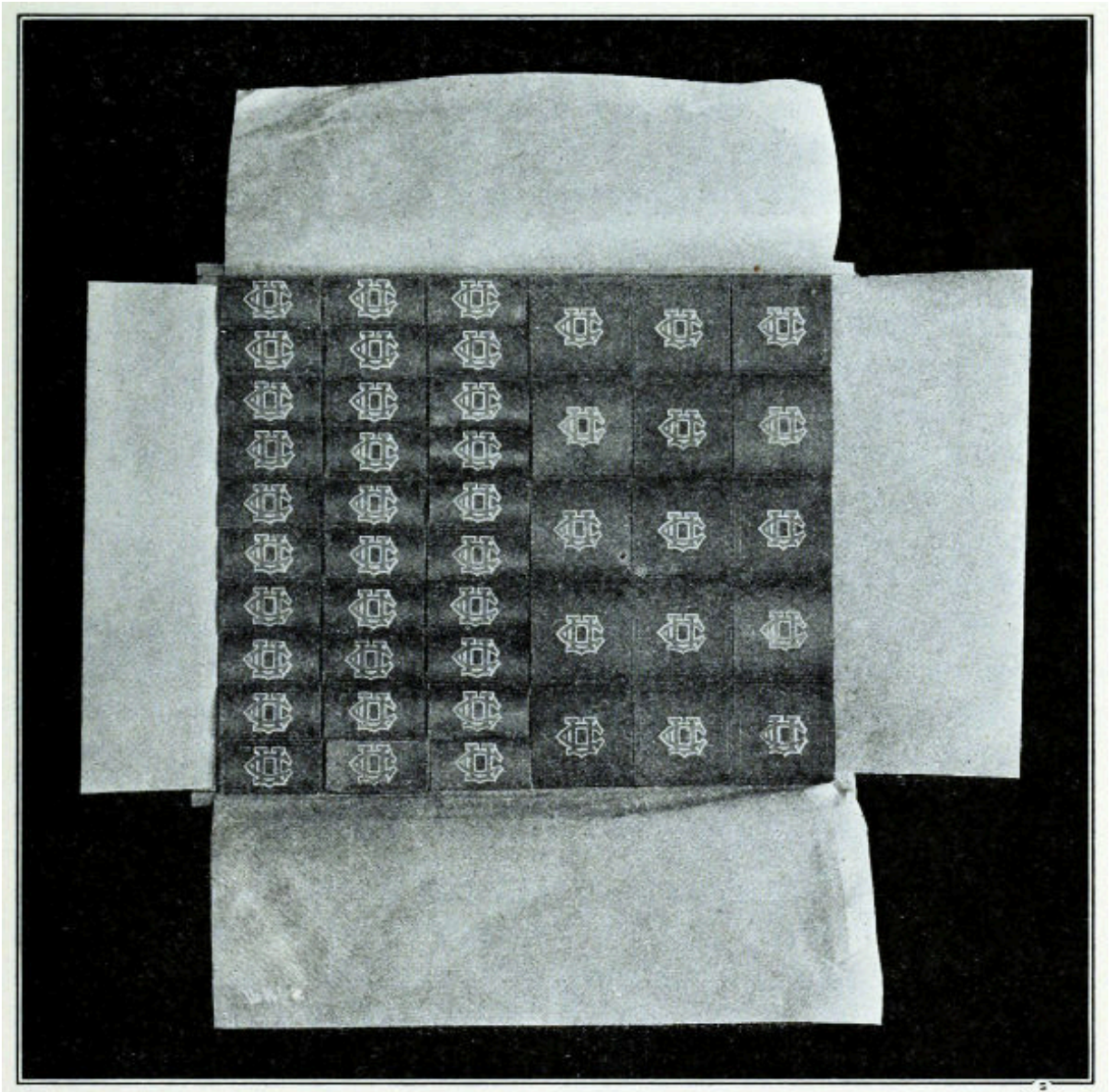
*** START OF THE PROJECT GUTENBERG EBOOK COMPARISON
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COMPARISON OF WOODS FOR BUTTER BOXES

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Butter boxes used in shipping and storing butter in California, are usually made of spruce which is largely shipped in from other states particularly from Washington and Oregon.

With the recent war, however, there came an acute shortage of spruce on the Pacific Coast with a corresponding increase in price. The commercial manufacturers did some work in an attempt to find a substitute for spruce, but the trade did not readily accept a change. There was a demand from both the lumber and the butter interests for investigation to find a suitable substitute for spruce.

The production of spruce is somewhat limited in California, but there is an abundance of white fir and a limited amount of cottonwood available. However, the creamerymen have not used white fir and cottonwood to any extent for butter containers, on account of the belief that these materials would impart a wood flavor to the butter.

Inasmuch as nearly all of the butter made in this State is shipped or stored in wooden containers, the use of white fir or cottonwood, would mean first, a material saving to the butter manufacturers in marketing expense, and second, an opportunity for the lumber interests to use a large amount of raw material already available in California, which heretofore had been of little commercial value or use.

BUTTER ABSORBS ODOR

The volatile fats in butter have the property of absorbing odors, which often results in an undesirable flavor. Great care then must be exercised in keeping butter from coming in contact with materials that will impart a foreign flavor. Butter need be exposed to foreign odors only a short length of time before the flavor is permanently affected.

Experiments^[1] were conducted, therefore, to determine whether white fir or cottonwood would impart a flavor to the butter and also to determine the possibility of storing butter in cubes and marketing it in 60-pound cases when these woods were used.

CUBE BUTTER IN COLD STORAGE

The butter for cold storage was packed in white fir, cottonwood, and spruce containers holding ten pounds each. Both seasoned and unseasoned woods were used in each of the three methods of packing.

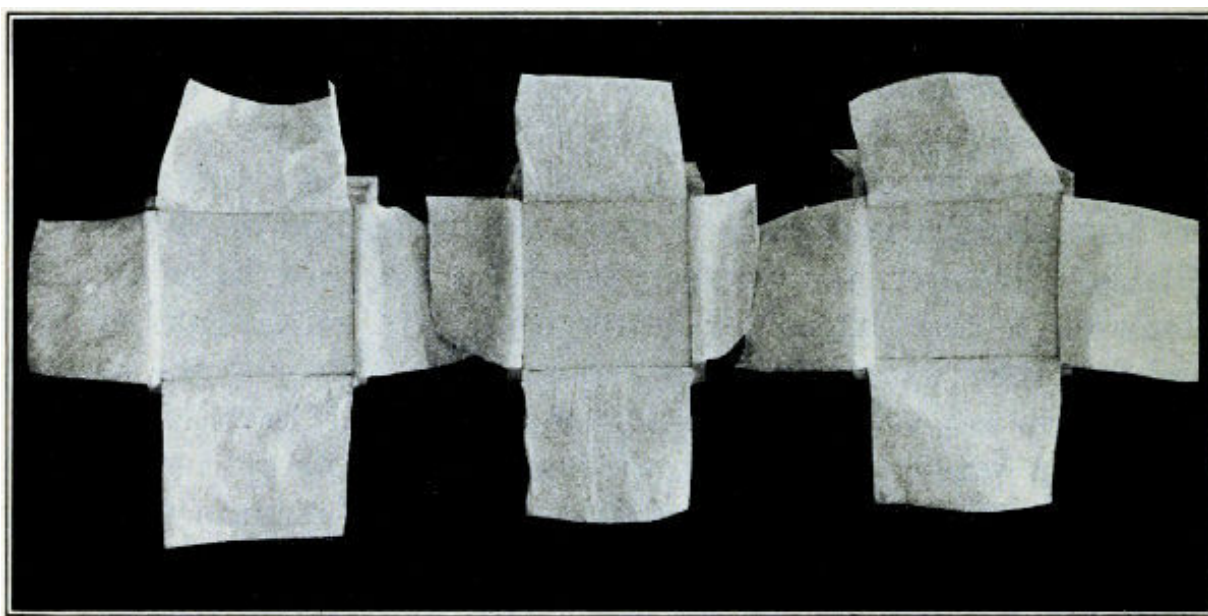


Fig. 1.—Butter packed in cubes paraffined and parchment lined.

The first set packed with butter were plain unseasoned boxes of each of the woods. The second set had the inner surface paraffined before packing. The method of paraffining was to invert the box over a steam jet and steam thoroughly. This served a double purpose in that it opened the pores of the wood and allowed the paraffin to penetrate, and the heated surface of the wood kept the paraffin in a liquid condition so that it could be put on in a

thinner coat than if the paraffin had been applied to a cold surface. After the boxes had been allowed to drain, the inside was then painted with paraffin at 240° F. This method gave a complete covering to the wood, a result which is not always obtained by some of the commercial paraffin atomizers. The third set was paraffined as above and, in addition, lined with good parchment paper so that no butter could come in contact with either wood or paraffin (fig. 1). Twenty-three 10-pound boxes were packed in the three ways.

They were filled with the butter from one churning which scored 92½ after being chilled for 24 hours at 50° F. and were shipped immediately after the first scoring to a cold storage plant in San Francisco and stored at a temperature of 12° F. The butter was scored monthly for six months. The summary of the scoring is given in [table 1](#).

TABLE 1
INFLUENCE OF VARIOUS WOODS ON CUBE BUTTER IN STORAGE [\[2\]](#)

No. of sample	Kind of wood	How treated	First score	Lowest score	Average of all scores	Average score of butter in same kind of box
1	White Fir	Unseasoned No Paraffin No Parchment	92.5	89	90.857	
2	Cottonwood	Unseasoned No Paraffin No Parchment	92.5	86	89.214	
3	Spruce	Unseasoned No Paraffin No Parchment	92.5	88	90.785	
4	Spruce	Seasoned Paraffin No Parchment	92.5	89	90.642	
6	White Fir	Seasoned Paraffin No Parchment	92.5	90	90.857	
7	Cottonwood	Seasoned Paraffin No Parchment	92.5	87	89.571	
5	Cottonwood	Unseasoned Paraffin No Parchment	92.5	88	89.857	
8	Spruce	Unseasoned Paraffin No Parchment	92.5	90	90.928	

9	White Fir	Unseasoned Paraffin No Parchment	92.5	89	90.571	
10	Cottonwood	Unseasoned Paraffin Parchment	92.5	91.0	91.714	
11	Cottonwood	Unseasoned Paraffin Parchment	92.5	89.0	90.571	91.142
18	Cottonwood	Unseasoned Paraffin Parchment	92.5	89.0	91.142	
12	Spruce	Unseasoned Paraffin Parchment	92.5	90.5	91.5	
13	Spruce	Unseasoned Paraffin Parchment	92.5	91.0	91.571	91.333
14	Spruce	Unseasoned Paraffin Parchment	92.5	90.0	90.928	
15	White Fir	Unseasoned Paraffin Parchment	92.5	89.0	90.928	91.107
16	White Fir	Unseasoned	92.5	90.0	91.285	
17	White Fir	Seasoned Paraffin Parchment	92.5	89.0	90.857	
20	White Fir	Seasoned Paraffin Parchment	92.5	90.5	91.428	91.142
19	Cottonwood	Seasoned Paraffin Parchment	92.5	90.5	91.571	
21	Spruce	Seasoned Paraffin Parchment	92.5	90.0	91.214	
22	Spruce	Seasoned Paraffin Parchment	92.5	90.0	91.571	91.523
23	Spruce	Seasoned Paraffin Parchment	92.5	91.0	91.785	

SIXTY-POUND BOXES PACKED FOR MARKET

The butter for market was cut into two-pound squares, wrapped and packed in 60-pound containers, made of white fir, cottonwood and spruce (figs. 2, 3 and 4). The butter was stored in a cold room, the temperature of which ranged from 48° to 50° F. It was held in storage twenty-eight days, which is within two days of the maximum time butter may be held and still sold as fresh butter. Butter held over thirty days must be labeled “storage butter.” The butter was scored four times during the storage period. The butter used was all from the same churning which scored 93 after being chilled for twenty-four hours at 50° F.

[Table 2](#) gives a summary of the scores showing the effect upon butter in containers with varying treatments. When paraffined, the inside of the boxes was painted with the paraffin at 240° F.

TABLE 2
INFLUENCE OF VARIOUS WOODS ON BUTTER PACKED IN 60-LB. BOXES
[\[3\]](#)

No. of sample	Kind of wood	How treated	Highest score	Lowest score	Average of all scores
1	White Fir	Unseasoned, Not Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
3	Cottonwood	Unseasoned, Not Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
7	Spruce	Unseasoned, Not Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
1-a	White Fir	Unseasoned, Not Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	—	—	—
3-a	Cottonwood	Unseasoned, Not Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	93	93
7-a	Spruce	Unseasoned, Not Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	93	93
2	White Fir	Seasoned, Not Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
9	Cottonwood	Seasoned, Not Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93

8	Spruce	Seasoned, Not Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
2-a	White Fir	Seasoned, Not Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	93	93
9-a	Cottonwood	Seasoned, Not Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	90	91.175
8-a	Spruce	Seasoned, Not Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	93	93
4	Cottonwood	Seasoned, Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
5	White Fir	Seasoned, Box Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
6	Spruce	Seasoned, Paraffined Parchment Wrapped, Cartons Box Lined with Wrapping Paper	93	93	93
4-a	Cottonwood	Seasoned, Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	93	93
5-a	White Fir	Seasoned, Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	92.75	92.562
6-a	Spruce	Seasoned, Paraffined Parchment Wrapped, No Cartons Box Lined with Parchment	93	93	93

NAILING OF BOXES

Five-penny cement-coated nails were used in making the boxes. Practically no splitting was caused by the nails in unseasoned white fir, spruce, or cottonwood. There was very little splitting in seasoned cottonwood. The nails, however, caused a slight splitting in the seasoned spruce and quite a noticeable splitting in the white fir, but not enough in either to cause an appreciable loss.

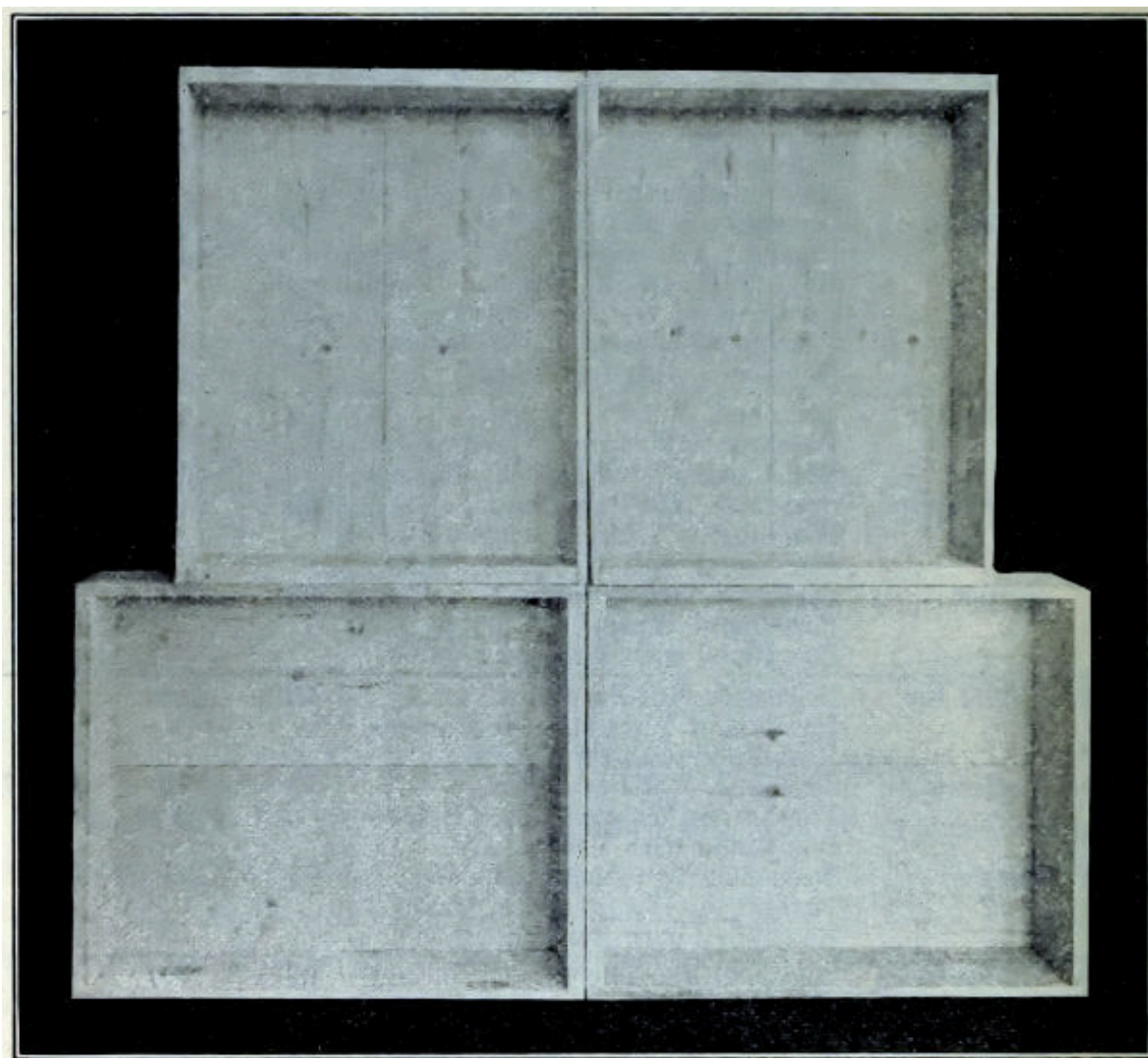


Fig. 2.—Typical 60-pound white fir boxes showing general run of this wood.

CONCLUSIONS

Cube Butter in Cold Storage

Boxes paraffined and parchment lined.—White fir and cottonwood can be used in place of spruce for storing butter in cubes, when properly seasoned, paraffined, and parchment lined.

Cottonwood is equal to spruce as a butter container. Butter stored in cottonwood boxes for six months had an average score of 0.048 of a point

above spruce treated in the same manner.

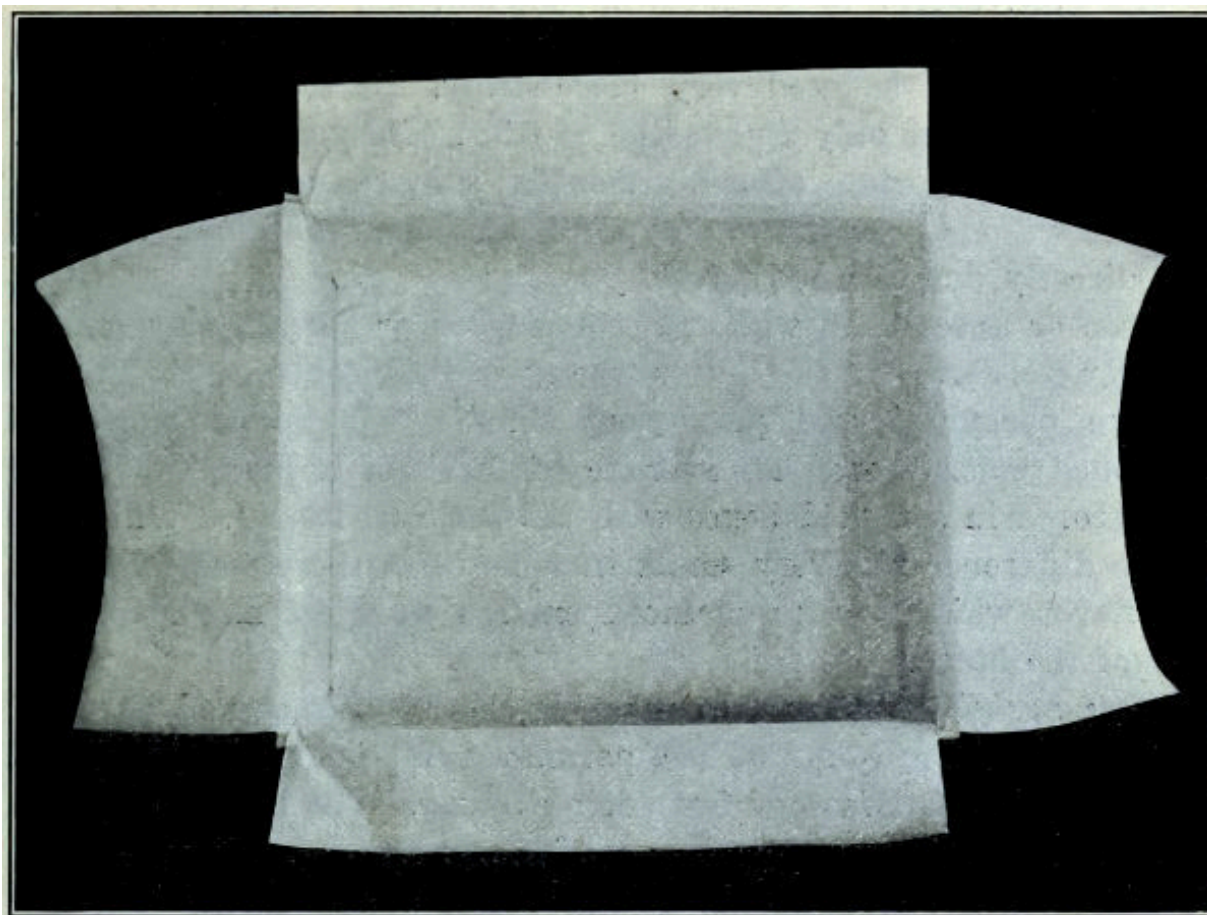


Fig. 3.—Method of lining 60-pound boxes with paper.

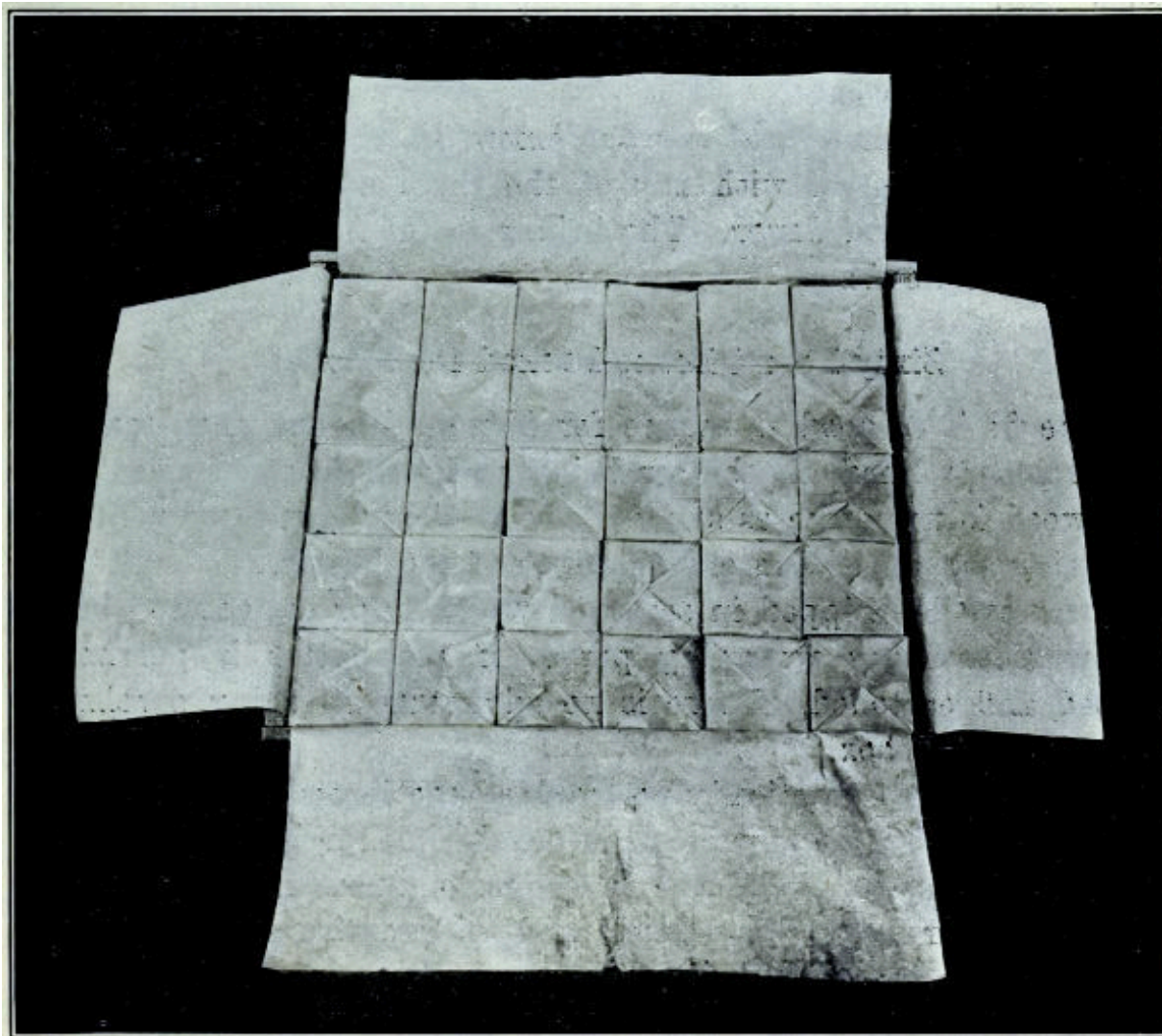


Fig. 4.—Parchment wrapped butter in parchment lined box.

White fir may be used very successfully. It scored during the six months' storage only an average of 0.381 of a point below spruce.

In the final scoring, after six months' storage, none of the cubes packed in seasoned, paraffined and parchment lined containers received a cut directly due to wood flavor.

Green or unseasoned white fir, cottonwood or spruce, may impart a slight wood flavor to the butter when packed in cubes, even though they are paraffined and parchment lined. The butter stored in unseasoned cubes scored an average of 0.218 of a point below the butter stored in seasoned boxes with the same treatment. While the average difference was very

small, in some cases there was a decided wood flavor which was pronounced enough to affect materially the flavor of the butter.

Boxes paraffined but not parchment lined.—Unseasoned boxes of white fir, cottonwood and spruce, paraffined but not parchment lined are not entirely satisfactory for storing butter. The butter so stored was criticized in practically all cases for wood flavor. Butter stored in white fir boxes scored 0.358 of a point lower than that in spruce boxes, while butter in cottonwood boxes scored 1.071 lower than that in spruce. Storing butter in cubes without parchment lining or in cubes carelessly lined with parchment will cause objectionable flavors regardless of the wood.

Boxes neither paraffined nor parchment lined.—Butter allowed to come in direct contact with any of the three untreated woods will always take up wood flavor. The injury to the flavor is about equal from all three woods.

Sixty-Pound Boxes Packed for Market

White fir is as good as spruce for 60-pound boxes when seasoned and parchment lined, the butter being wrapped in parchment only. Cottonwood is not quite as satisfactory as either spruce or white fir, there being some criticism on the flavor of the butter.

Butter can be shipped in seasoned white fir or cottonwood boxes, lined with ordinary wrapping paper, if the butter is parchment wrapped and cartoned. There is no advantage in using parchment paper to line the box.

Since there was practically no trouble experienced in the unparaffined boxes, there is no advantage in paraffining the inside of the box.

Since the completion of the investigational work, approximately 40,000 white fir boxes have been used with entire satisfaction for shipping butter at the University Farm.

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[1] This experiment was suggested by Mr. M. B. Pratt, Deputy State Forester. Through his coöperation, all box material was furnished by the Swayne Lumber Company of Oroville and the Capitol Box Factory of Sacramento.

[2] This scoring was done by T. J. Harris, San Francisco Dairy Produce Exchange, S. L. Denning, Oakland, and G. D. Turnbow, College of Agriculture, University of California.

[3] Butter scored by J. C. Marquardt and G. D. Turnbow of the College of Agriculture, University of California.

Transcriber's notes:

In the text version, italics are represented by underscores, and bold text by =equals= symbols.

The bulletins and circulars sections have been expanded from 2 columns in small font to a single column to allow them to be more easily read.

The single occurrence of paraffine has been changed to paraffin for consistency with general use in the text.

***** END OF THE PROJECT GUTENBERG EBOOK COMPARISON OF
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